

REPORT 2.2 — Testing verdict on build v24 before it is played (Fable, 2026-09-14)

Item: docs/handoff/FOR_FABLE.md 2.2. Tree tested: 0971d7b (clean, 19 commits ahead of the phone's 0ba7721 / v23). Effort high. His words on it:

"Go ahead and I'll take your recommendations. We need the lock button bigger and available at all times"

"I hit a tree off the tee on hole 1 and thought I had marked the shot but I think you made selecting a lie required so I couldn't figure out why I couldn't mark my 3rd shot at the ball on 1 so I ended up deleting the shot marking twice"

"Then I had to mess with marking the shot and the cup, trying to lock the phone because you cant lock it on the putting screen"

The answer, in plain words

Do not put v24 on the phone yet. Keep playing v23.

The lock is right: the LOCK tab is on screen over every sheet, including the putting sheet, nothing tappable sits under it on any sheet, and the change to the unlock gesture does not weaken the pocket guard. The marks and the round save are right: every shot, the cup and the finished round landed in storage exactly as tapped, locked or not.

What is wrong is the lie question. In v24 it moved from the footer into a card at the top of the screen, above the action buttons that never change. On a phone the size of yours (360 x 780) that card does not fit: at the moment a shot is saved, the first row of lies (TEE / FAIRWAY / ROUGH) is 39 px below the bottom of the scrolling area and the second row (SAND / RECOVERY / GREEN) is off screen entirely. You would have to scroll to answer the lie - the same "required thing I cannot find" that cost you two deleted marks on hole 1. In v23 the lie panel was always fully on screen.

I fixed one thing myself: the card was also 30 px too wide (its grids grew past the body), so the club chips "5" and "PW" sat under the LOCK tab at 375 px and the whole right-hand column at 360 px - a thumb going for a lie or a club would have locked the phone. That is committed with four tests. The fit above the fold is Opus's screen to rework; two tests that say exactly what must be true are left RED in the suite as the bar, with the numbers below.

Verdict

FAIL for the course, on one defect: the lie grid is below the fold at burst end at a phone's height. Everything else asked passes:

asked	result
lock tab above every sheet, nothing tappable under it	PASS - 11 sheet kinds + 3 confirm sheets at 375 x 812 and 8 at 360 x 780, 22 px clearance on every one, 0 buttons under the tab
the marks	PASS - tee, shots 2-5, cup under LOCK; all in localStorage as tapped
round save	PASS - gaps gate, finish confirm, status completed, 5 shots + cup intact, key present after a page load
zoneOf review	PASS - does not weaken the guard (Section 5)
"known intermittent" diagnosis	CONFIRMED (Section 4) - the caveat is dropped from CLAUDE.md and .claude/agents/fable.md
nothing tappable under the tab on the play screen	FAILED in v24 as delivered, FIXED here (Section 6)
lie grid on screen at burst end	FAILED, open for Opus (Section 7)

1. What was read

git show 84da7f4 in full (js/ui/lock.js, css/base.css, js/ui/screen-play.js, sw.js, js/data/build.js, test/run.js, test/index.html, the v24 section of docs/REVISIONS.md); git diff --stat 84da7f4..0971d7b -- js css sw.js test index.html = only js/data/courses.js (the hole 7/10 yardages) and the matching test - nothing else in the web app moved after v24. js/ui/lock.js at HEAD in full; js/ui/dom.js (sheet, confirmSheet, toast); the thirteen sheet(callers in js/ui/screen-play.js and js/ui/screen-history.js; the sheet, scrim, lock-tab, lock-screen and toast rules in css/base.css; js/app.js lines 151-193 (idleMs, canLock, activity wiring).

2. Suite (<http://localhost:8123/test/,tools/devserver.py> 8123)

Browser pane HIDDEN for every run (`document.visibilityState "hidden"`).

run	tree	result
1, 2, 3	0971d7b untouched, no viewport emulation (<code>window.innerHeight 0</code>)	501 / 501 each; "the deliberate gesture unlocks" passed all three
4	mutation: zoneOf back to <code>window.innerHeight</code> , no emulation	500 / 501 - exactly "the deliberate gesture unlocks": unlocked: expected false, got true
5	new strip tests, CSS fix NOT applied (375 only; the 360 case had a harness fault, fixed)	501 / 505 - all four strip tests fail: "CANCEL" ends at 306 px, strip starts at 289
6	strip tests + CSS fix	505 / 505
7	strip tests, CSS fix stashed out again (proof at both widths)	501 / 505 - "CANCEL" ends at 307 px, strip starts at 289 (375) and ... starts at 274 (360)
8, 9, 10	final tree (this commit)	505 / 509 each - the four RED are the acceptance tests of Section 7, named below

The four RED on the final tree, on purpose (the label numbers move by 1 px between runs with the emulated size):

- at 375 px every label in the lie card fits its button - "RECOVERY" is 5-6 px wider than its button
- at 375x812 the whole lie grid is above the footer when the burst ends - lowest lie button ends at 428 px, the body at 382 px (46 px below the fold)
- at 360 px every label in the lie card fits its button - "RECOVERY" is 10-11 px wider than its button
- at 360x780 the whole lie grid is above the footer when the burst ends - lowest lie button ends at 428 px, the body at 350 px (78 px below the fold)

"The deliberate gesture unlocks" passed on all nine non-mutation runs.

3. What was clicked and read back (?sim=1, one tab)

375 x 812 unless stated; club tracking on (the default). Scripted clicks on the real buttons, DOM and localStorage read back after each step. Every sheet was measured with the round live and unlocked: the tab's rect, every button / input / select / [role=button] / .seg-btn / .list-row in the sheet, the right-most one level with the tab, any element whose rect intersects the tab's, and `elementFromPoint` across the tab's face on an 8 px lattice.

Sheets at 375 x 812 (n = 11 kinds + 3 confirm sheets): Hole 1 - putts (75 tappables), Hole 1 - yardages (8), Penalty (9), Shot 2 editor (21), Go to hole (19), Round menu (13), Hole 1 - hand entry (9), Hole 1 - how did it go? (28), Hole 1 - confirm your shots (4), Missing data (3), plus the confirm sheets Abandon round? (2), Finish round? (2) and Cup at the tee? (2). Every one: padding-right 98 px, right-most button ends at x = 277, tab starts at x = 299, **22 px clearance, 0 under the tab**. `elementFromPoint` on the tab's face: 208 of 210 lattice points hit the tab; the 2 misses are the rounded corners at (301, 389) and (301, 549) and land on .footer, which is not tappable. Not opened: Starting hole (needs learned tees to fire; same .sheet rule, so the same padding) and the history sheet (no round is live there, so there is no tab). The Shot 2 editor offers no distance button off a par 3, so the distance sheet was not reached (it is a sheet() like the rest).

Sheets at 360 x 780 (n = 8): putts, yardages, Penalty, Go to hole, Round, hand entry, how did it go?, Shot 2 editor - right-most button at x = 262, tab at x = 284, 22 px clearance, 0 under.

Toast: fixed at the bottom, 730-798 px, right edge 363; tab ends at 555. No overlap at any height a phone has; z-index 60 under the tab's 70.

Settings screen mid-round (the tab is on every screen): the "Show scoring and distances" segmented control (NEVER / TOURNAMENT / ALWAYS) is 236 px wide in a 250 px field, but its columns are content-sized: "ALWAYS" ends at x = 331, and scrolled level with the tab it is **32 px under it**. Same root cause as Section 6, pre-dates v24 (the v23 strip was 16 px narrower, so it was under by 16 then). Off the course path; handed to Opus, not fixed - it is his screen and the fix is a layout choice (columns on segmented, or a shorter label).

Marks. MARK TEE SHOT (card data-burst="running" in the body, footer "MARK TEE SHOT [disabled]", the other five buttons live) -> "Tee shot marked.", stored seq 1, lie tee, method burst. MARK SHOT 2 -> card done in the body with lie grid, club grid, CANCEL SHOT / LIE LATER; footer "MARK SHOT 3" enabled; stored seq 2, lie

fairway, lieInferred true; ROUGH -> lie rough, no lieInferred, card gone. Shots 3-5 the same way (GREEN on shot 4 opened the putts sheet on its own, as coded). MARK CUP with LOCK at 0.4 s: burst ended under the overlay; the sim is static, so "Cup at the tee?" was waiting under the lock (.scrim z 50, .lock-screen z 200), cup not yet stored; unlock -> MARK CUP HERE -> cup.method burst, 5 fixes used, footer "RE-MARK CUP", putts sheet reopened.

Auto-lock. It fired on its own 30 s into the first session: scripted .click() calls are not pointer activity, so the idle timer ran out with the sheets open - the app doing what it is meant to. (A script click passes through the overlay; a finger cannot.) For the rest of the measurements idleMs was raised to 600 s through configure(), in the page only.

Round save. Round menu -> Finish round -> "Missing data: 2 things are missing ... Hole 1: lie for shot 3 not chosen / Hole 1: no putts entered" (both true: LIE LATER on shot 3, and my script hit the putts sheet's GPS button instead of SAVE) -> SAVE WITH GAPS ANYWAY -> "Finish round? 17 holes were never started" -> FINISH -> summary. localStorage: status completed, completedAt set, activeRoundId null, revision 4, simulated true, hole 1: 5 shots with their lies as tapped, cup present. The lock disabled itself and the tab left with it (has-lock-tab off). On the next page load the round's key was still there (64 gt:round: keys in this dev browser). n = 1 round, 1 hole, 5 shots, 1 cup.

4. The "known intermittent" - confirmed, and why it looked intermittent

With the pane hidden and no viewport emulation on the tab, the test page reports window.innerHeight 0 (read directly, three runs). With zoneOf measuring window.innerHeight the "top" test is $y < 0$ - never true - so every tap reads as "bottom" and step 1 can never be taken: "the deliberate gesture unlocks" fails every time (run 4, the mutation). With zoneOf measuring the overlay it passes every time (runs 1-3, 6, 8-10). Deterministic both ways, n = 10 runs.

The extra piece Opus did not have: **with a viewport emulation set on the tab (375 x 812, 360 x 780), a hidden pane reports the emulated height**, not 0 (read on runs 5-10). So the old test passed whenever the session had an emulated size on the tab and failed whenever it did not - which is why it came and went between sessions and reports. Caveat dropped from CLAUDE.md Section 4 and .claude/agents/fable.md.

5. zoneOf - critical review

The whole code change in js/ui/lock.js is one line (git show 84da7f4 -- js/ui/lock.js):

```
const h = el.getBoundingClientRect().height || window.innerHeight;
```

- **Same number in production.** .lock-screen is position: fixed; inset: 0 (css/base.css), so its height IS the viewport the tap's clientY is measured in. When the two ever differ (browser chrome, an on-screen keyboard), the overlay is the honest one: the taps are read off it.
- **Fallback is the old behaviour.** || window.innerHeight only when the overlay measures 0, which is exactly what pre-v24 used.
- **Dead band, split, rejections untouched.** TIMING.deadBand 0.14, top is $y < h * 0.43$, bottom is $y > h * 0.57$; one contact at a time, tap ≤ 350 ms and ≤ 24 px, top then bottom, both inside 1200 ms, every rejection resets to step 1 - all read in the function at HEAD, none in the diff.
- **Failure directions.** If h were ever smaller than the screen the bottom zone grows, but the first tap must still be a crisp single contact in the top 43% - no single contact unlocks. If h were larger, the bottom zone shrinks or leaves the screen and unlocking gets harder, never easier. The one nit: y is not offset by the overlay's top; correct because inset: 0 pins it to 0, and worth a line if the overlay ever moves.

Verdict: it does not weaken the pocket guard. Approved.

6. Defect 1 - the card grew into the lock strip (fixed here)

The lie card is a grid inside .body, whose padding-right reserves the tab's strip (12 + 86 = 98 px), so the body's content ends at $x = 277$ (375) / 262 (360) and the tab starts at 299 / 284. But a grid column is minmax(auto, 1fr): labels wider than their share widen the column and the card grew to its contents - **x = 307 at both widths**.

width	v24 as delivered	after the fix
375 x 812	card right 307; lie grid right 292; club chips "5" and "PW" right 305 = 6 px under the tab ; elementFromPoint there returns the tab	card right 277; right-most button 271 (CANCEL / LIE LATER); 0 under
360 x 780	card right 307 vs tab at 284 = 23 px under : GREEN, LIE LATER, CANCEL and the right club column	card right 262; right-most button 255; 0 under

The tab is on top (z 70), so the consequence was an accidental LOCK on a lie or club tap - the shot is saved by then, so a lock and an unlock, not data. Still the rule the tab sits above the sheets on.

Fix (css/base.css, the .body > .capture block): min-width: 0; max-width: 100% on the card, repeat(3, minmax(0, 1fr)) on the lie grid, repeat(5, minmax(0, 1fr)) on the club grid, card side padding 12 -> 6 px, lie labels 16 -> 14 px with no tracking, club chips padding-inline: 2px. The columns are pinned by construction now, whatever the labels measure. Three lie columns kept (two would put RECOVERY and GREEN in a third row, 74 px lower).

Tests (test/run.js, group "the capture card stays out of the lock strip", 4 of its 8): the real play screen mounted at 375 x 812 and 360 x 780 with the real CSS, driven to the running card and the pending-lie card, every button's right edge <= width - gutter, and nothing past the card's own edge. Proven to bite: 4 / 4 fail with the fix stashed out (run 7).

7. Defect 2 - the lie grid is below the fold at burst end (open, Opus)

.body is the only scroller; the footer is a constant 430 px at 375 x 812 and 465 px at 360 x 780 (hint + MARK SHOT 88 + three 60 px buttons + the PENALTY / UNDO row + gaps). Above it: hud 61 + hole nav 70. So the body's visible box is 131-382 (251 px) at 375 x 812 and 131-315 (184 px) at 360 x

780. The card, top to bottom: head 156-198 (two lines now that the card is

its true width), bar 206-216, the boxed required field from 224 with its label, then the lie grid **288-428** (two rows of 66), then the club grid to 653, then CANCEL SHOT / LIE LATER to 710.

size	lie row 1	lie row 2 (SAND / RECOVERY / GREEN)
375 x 812, v24 as delivered	fully visible	26 px below the fold (40 of 66 visible)
375 x 812, after Section 6	fully visible	46 px below the fold
360 x 780	39 px below the fold (27 of 66 visible)	entirely off screen

Opus's "the lie box is fully visible (top 267, bottom 407 of an 812 px viewport)" measured against the viewport, not the body's bottom edge (382). In v23 the lie panel took the footer over and was always fully on screen; v24 trades that for a footer that never changes, which he asked for - but the two do not both fit in 184 px, and RECOVERY does not fit a 64 px column at any readable size either (the field box's 12 px padding and 3 px border and the card's padding take 36 px of the row).

This is his screen and what is on it at burst end is a course call, so it is not fixed here. The bar is in the suite, layout-agnostic: every label fits its button, and the lowest lie button ends above .body's bottom edge at burst end, at 375 x 812 and 360 x 780, club tracking on. Things that would move the numbers, for Opus to weigh: the lie grid first in the card with the head and bar under it (about 66 px); the required-field box's padding (24 px of width, 30 px of height); the club grid behind a single CLUB toggle (202 px of height, and it is optional); lie buttons 66 -> 56 px (20 px). The 30 px of body padding and the 465 px footer are the rest of the budget.

8. Decisions made alone

1. **Fixed the strip overflow in css/base.css myself.** It is a design-rule violation on the course path (a lie tap locking the phone), the fix is pinning grid columns, and a BLOCKED for it would have cost a spawn for a question with one answer. Logged.
2. **Three lie columns at 14 px, not two at 16 px.** Two columns fit the labels but put the third row 74 px lower on a card that is already below the fold. Both fail the fold; three columns fail it less.
3. **Left the fold to Opus and committed its acceptance tests RED.** What he sees at burst end on the course is his call through Opus's chat, not mine; a test proven to fail against the defect is the standard here, and "fully visible" was claimed once already by reading the wrong edge.

4. **Did not fix the Settings segmented control** (32 px under the tab when scrolled level). Same cause, off the course path, a label/columns choice in Opus's screen. Named for him.
5. **No BUILD.id bump**. v24 has never been deployed; this CSS rides in it. The bump to v25 belongs with Opus's rework, when a build is actually about to ship.
6. **Raised idleMs to 600 s in the sim page for the measurements** after the 30 s auto-lock fired under scripted clicks. Page-local, nothing stored.
7. **Dropped the intermittent caveat** from CLAUDE.md and .claude/agents/fable.md, as the item and my standing instructions say to once confirmed; replaced with the RED-on-purpose rule for the two fold tests.

9. For Matt to decide

Nothing on this item. v24 stays off the phone; the walking 9 this week goes with v23 unless Opus's rework clears the two RED tests and it is tested again.